

EC Communication Lab - Short Note + Diagrams

WITH LOVE, by KTULABS

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1. BASK Modulation

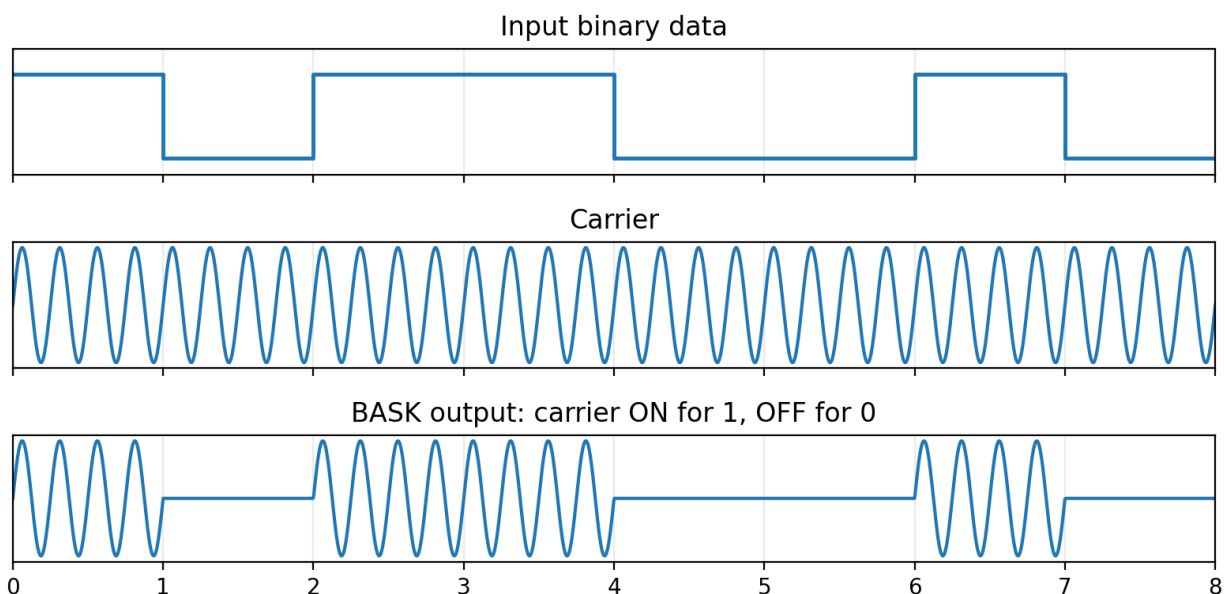
Aim: To generate Binary Amplitude Shift Keying, where carrier amplitude changes according to binary input.

Core idea: 1 -> carrier present, 0 -> carrier absent/low amplitude.

Block/working: Input data -> product/modulator with carrier -> BASK signal. Demodulation uses envelope/product detector + filter + comparator.

Procedure: Use a square wave as binary input. Apply sinusoidal carrier. Observe input bits, carrier, and BASK output on CRO/scope.

Clean diagram / waveform:



Notebook reference pages: 5, 6

2. QPSK Modulation

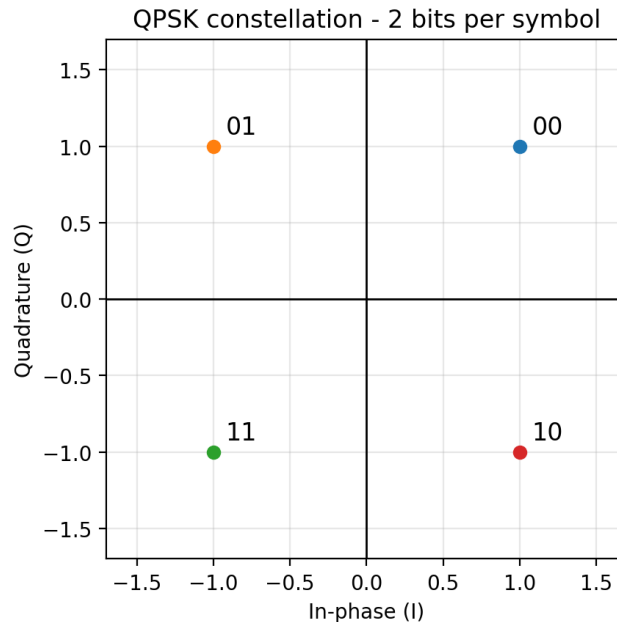
Aim: To generate Quadrature Phase Shift Keying, where each symbol represents two bits.

Core idea: Two carriers are used: In-phase carrier and 90 degree shifted quadrature carrier. Each dibit changes phase.

Block/working: Serial bits -> dibit encoder -> I and Q branches -> balanced modulators -> adder -> QPSK output.

Procedure: Group data into dibits. Map dibits to phase positions. Modulate I and Q carriers and add them. Observe modulated waveform/constellation.

Clean diagram / waveform:



Notebook reference pages: 18, 19, 20

3. BFSK Modulation

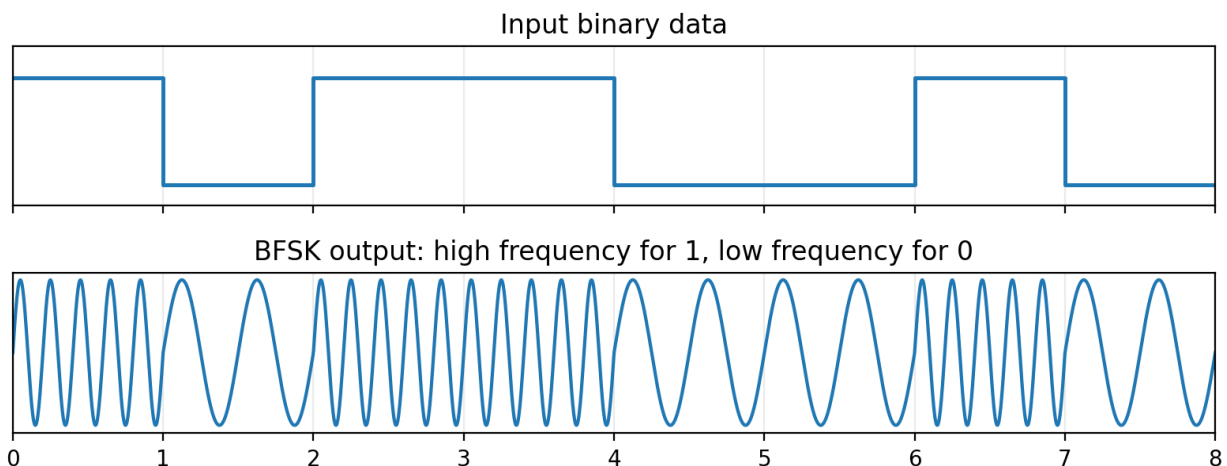
Aim: To generate Binary Frequency Shift Keying, where frequency changes according to input bits.

Core idea: 1 -> high frequency carrier, 0 -> low frequency carrier. Amplitude remains almost constant.

Block/working: Input bits select between two oscillators/carriers f_1 and f_2 . Output is continuous waveform with two frequencies.

Procedure: Apply binary data. Generate/choose two carrier frequencies. Switch carrier according to bit value. Observe data and BFSK waveform.

Clean diagram / waveform:



Notebook reference pages: 9, 10, 11, 12

4. Performance of Waveform Coding using PCM

Aim: To study Pulse Code Modulation and observe sampling, quantization and encoding.

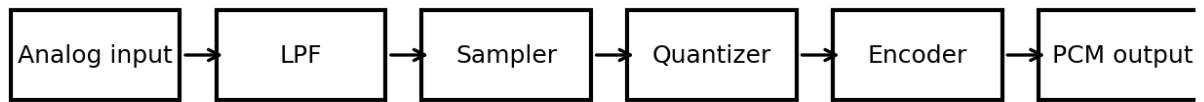
Core idea: Analog signal is converted into digital pulses using sampling -> quantization -> binary encoding.

Block/working: Analog signal -> LPF -> sampler -> quantizer -> encoder -> PCM output. Receiver reverses this using decoder and reconstruction filter.

Procedure: Generate sine/message signal. Sample it. Quantize samples into levels. Encode each level as binary PCM code. Compare reconstructed signal with input.

Clean diagram / waveform:

PCM transmitter block diagram



Notebook reference pages: 13, 14

5. BFSK Modulator / Circuit Reference

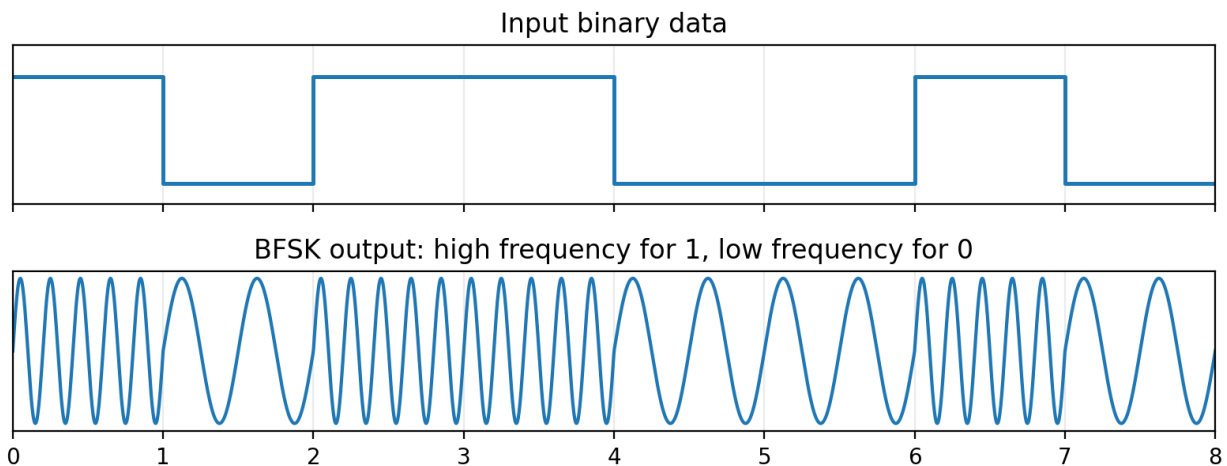
Aim: To study practical BFSK modulation circuit/block implementation.

Core idea: The modulator switches between two carrier frequencies based on binary input.

Block/working: Binary input controls the selection of f1 or f2 carrier. The selected carrier becomes BFSK output.

Procedure: Connect circuit/block as per diagram. Give digital input and observe carrier frequency shift in output.

Clean diagram / waveform:



Notebook reference pages: 9, 10

6. Generation of BFSK Signal

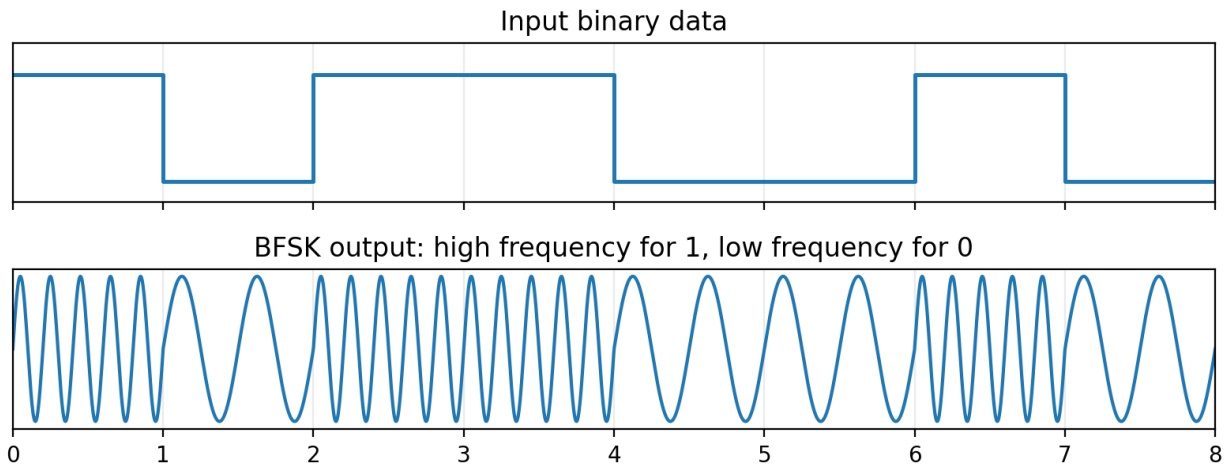
Aim: To generate a BFSK waveform using software/MATLAB and verify the output.

Core idea: The output frequency is changed based on the data bit stream.

Block/working: Define bit sequence -> define f1 and f2 -> generate carrier for each bit interval -> concatenate output.

Procedure: Enter bit sequence and frequency values. Plot input data and BFSK modulated signal. Check high/low frequency sections.

Clean diagram / waveform:



Notebook reference pages: 7, 8

7. Delta Modulation

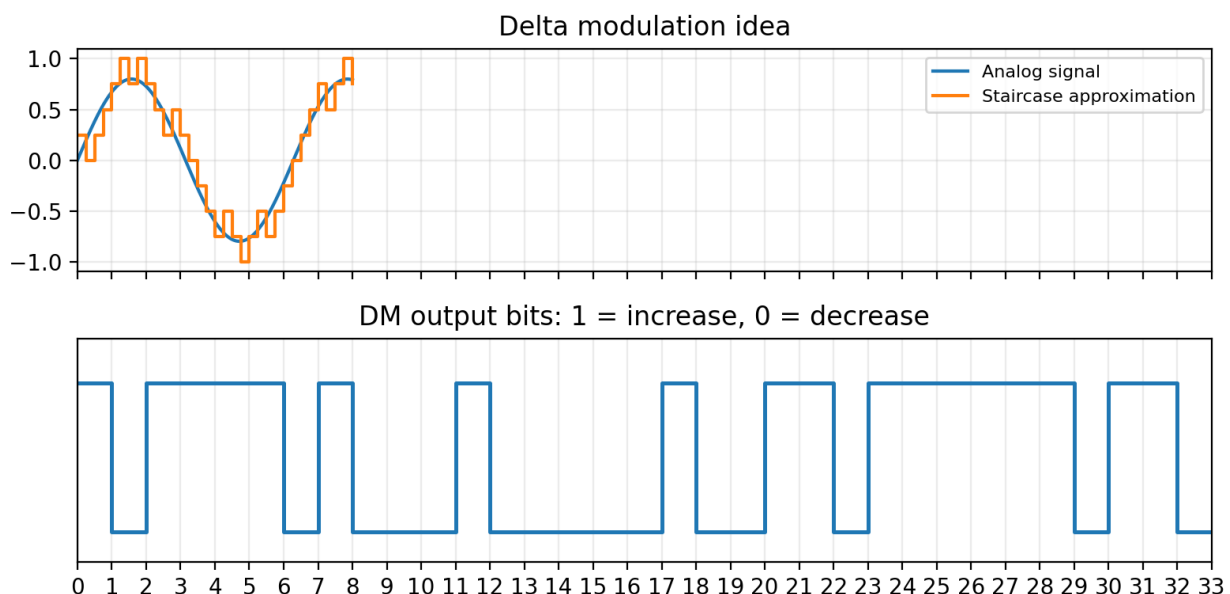
Aim: To study Delta Modulation, where only the change in signal is transmitted.

Core idea: 1 means signal is increasing; 0 means signal is decreasing. It uses staircase approximation.

Block/working: Input signal -> comparator -> 1-bit quantizer -> accumulator/integrator feedback -> DM output bits.

Procedure: Compare input signal with staircase approximation. If input is higher, send 1 and step up; else send 0 and step down.

Clean diagram / waveform:



Notebook reference pages: 15, 16, 17

8. Familiarization of Software Defined Radio

Aim: To understand SDR blocks and how radio functions are implemented using software.

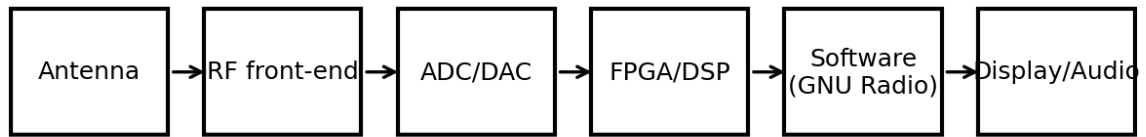
Core idea: In SDR, modulation/demodulation/filtering can be done in software instead of fixed hardware.

Block/working: Antenna -> RF front-end -> ADC/DAC -> DSP/FPGA -> software tools like GNU Radio -> output/analysis.

Procedure: Identify transmitter and receiver blocks. Observe how GNU Radio/software blocks replace traditional hardware stages.

Clean diagram / waveform:

Software Defined Radio basic block diagram



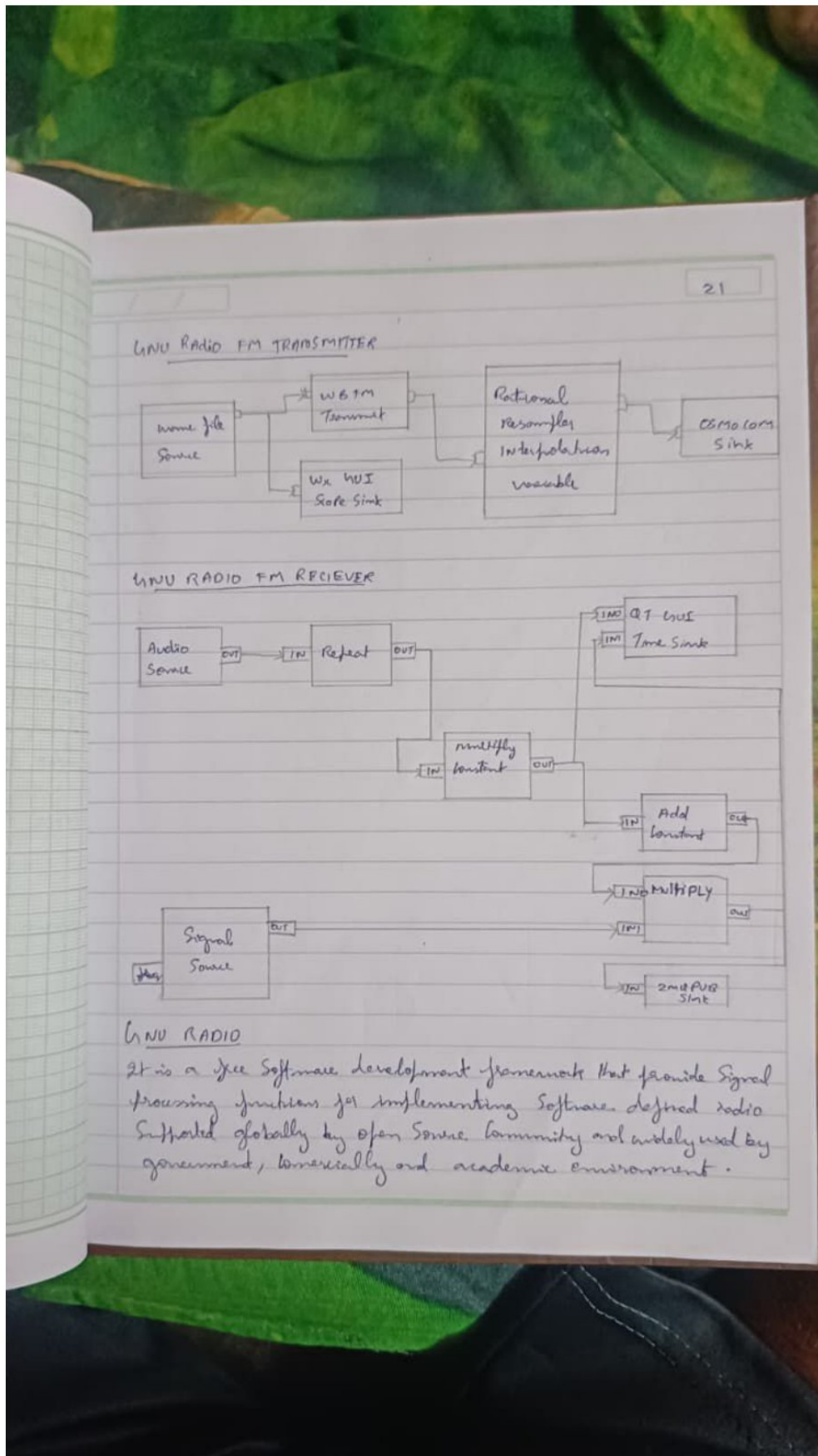
Main idea: hardware captures RF; software performs modulation/demodulation and analysis.

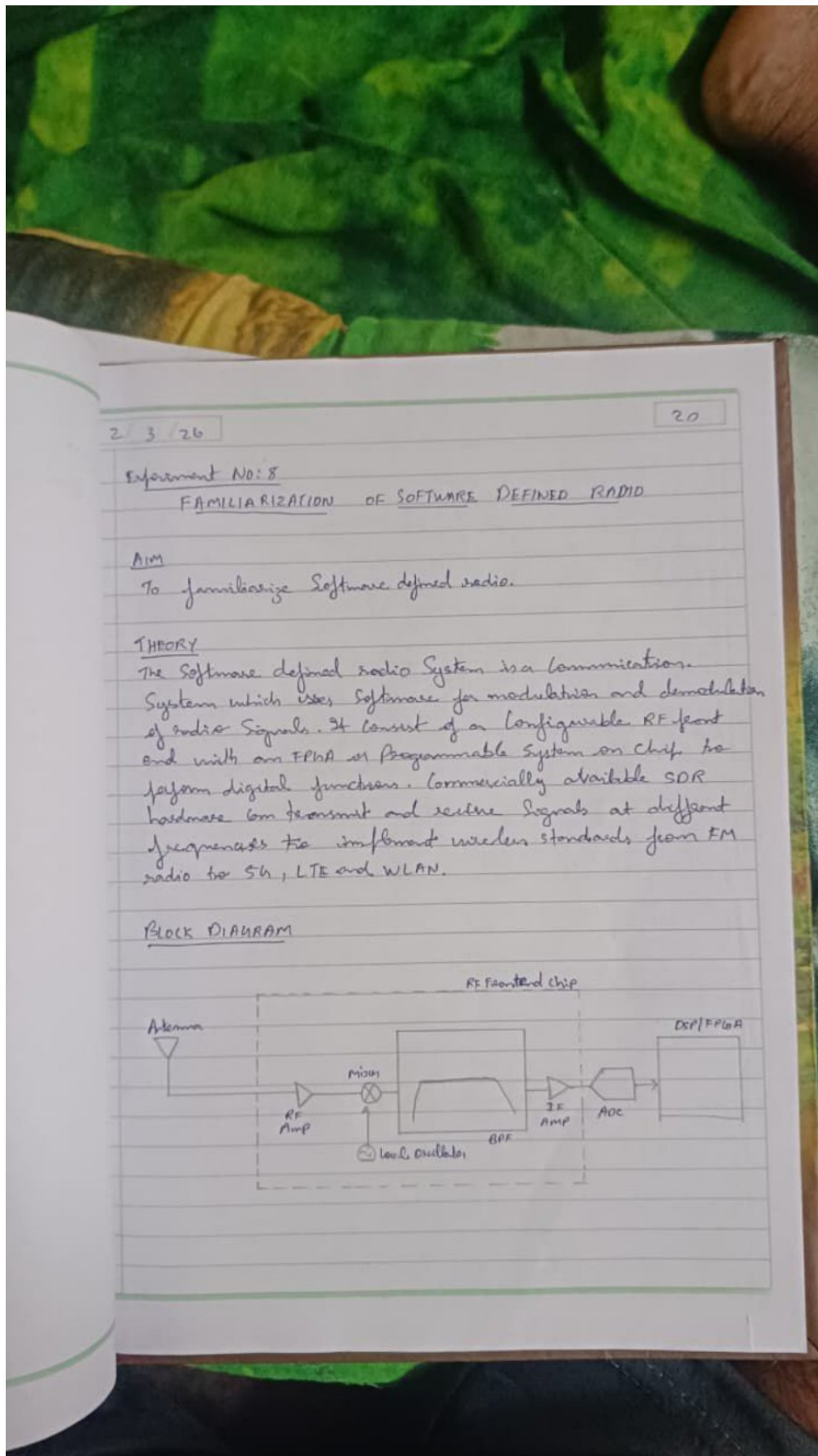
Notebook reference pages: 1, 2, 3

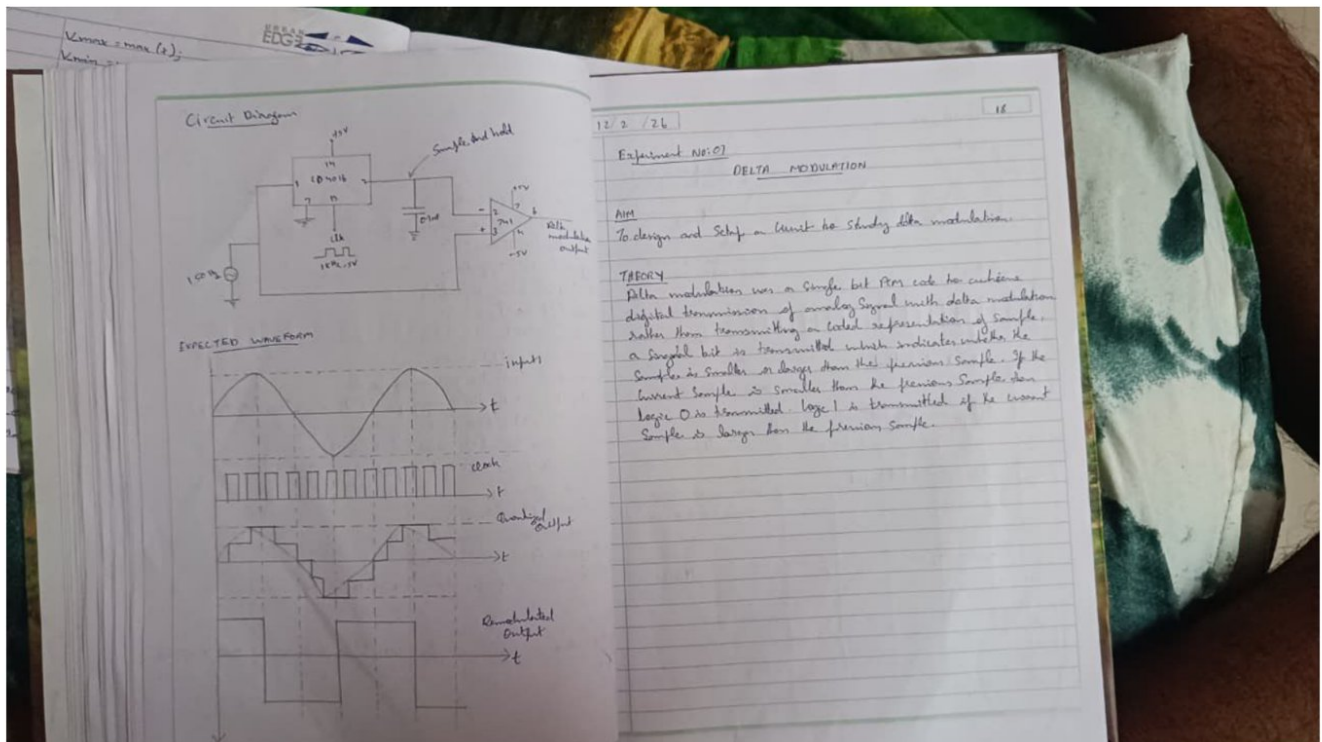
Original Notebook Diagram Pages

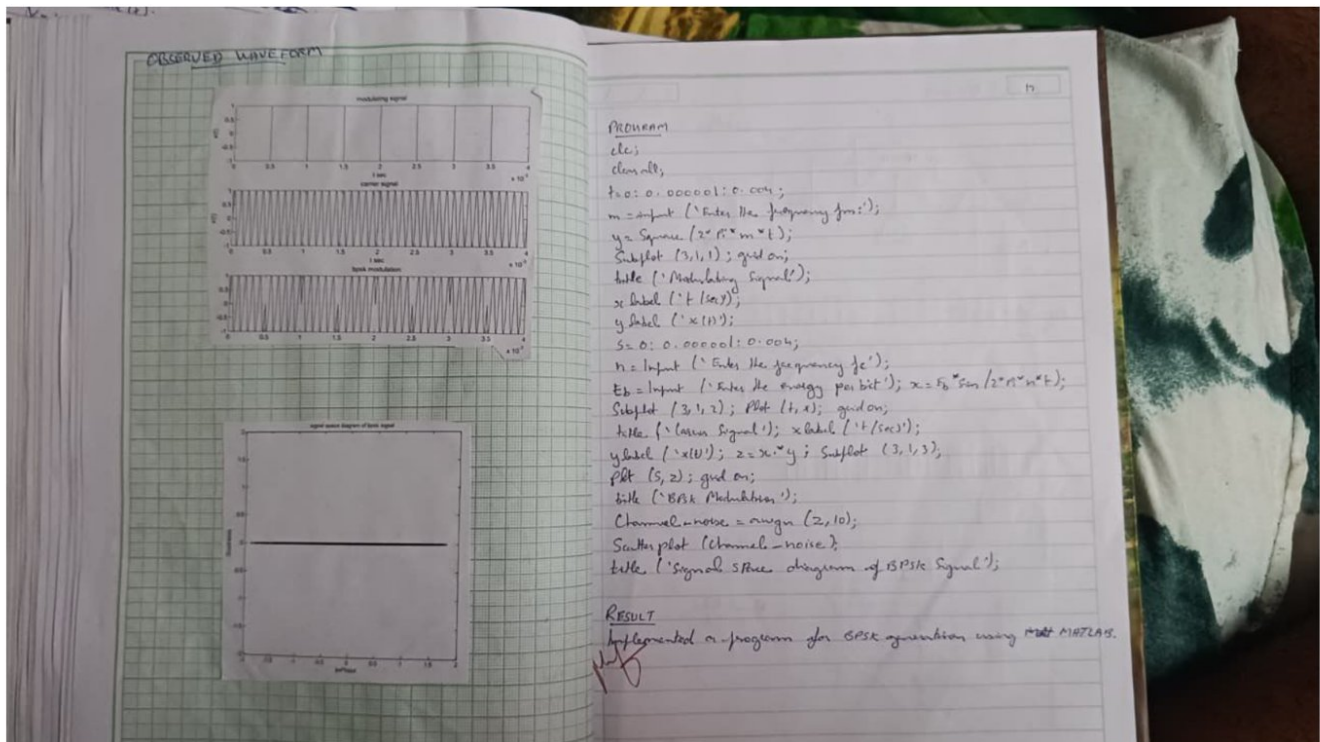
These pages are included so you can copy the actual lab-book drawings/waveforms when writing the record.

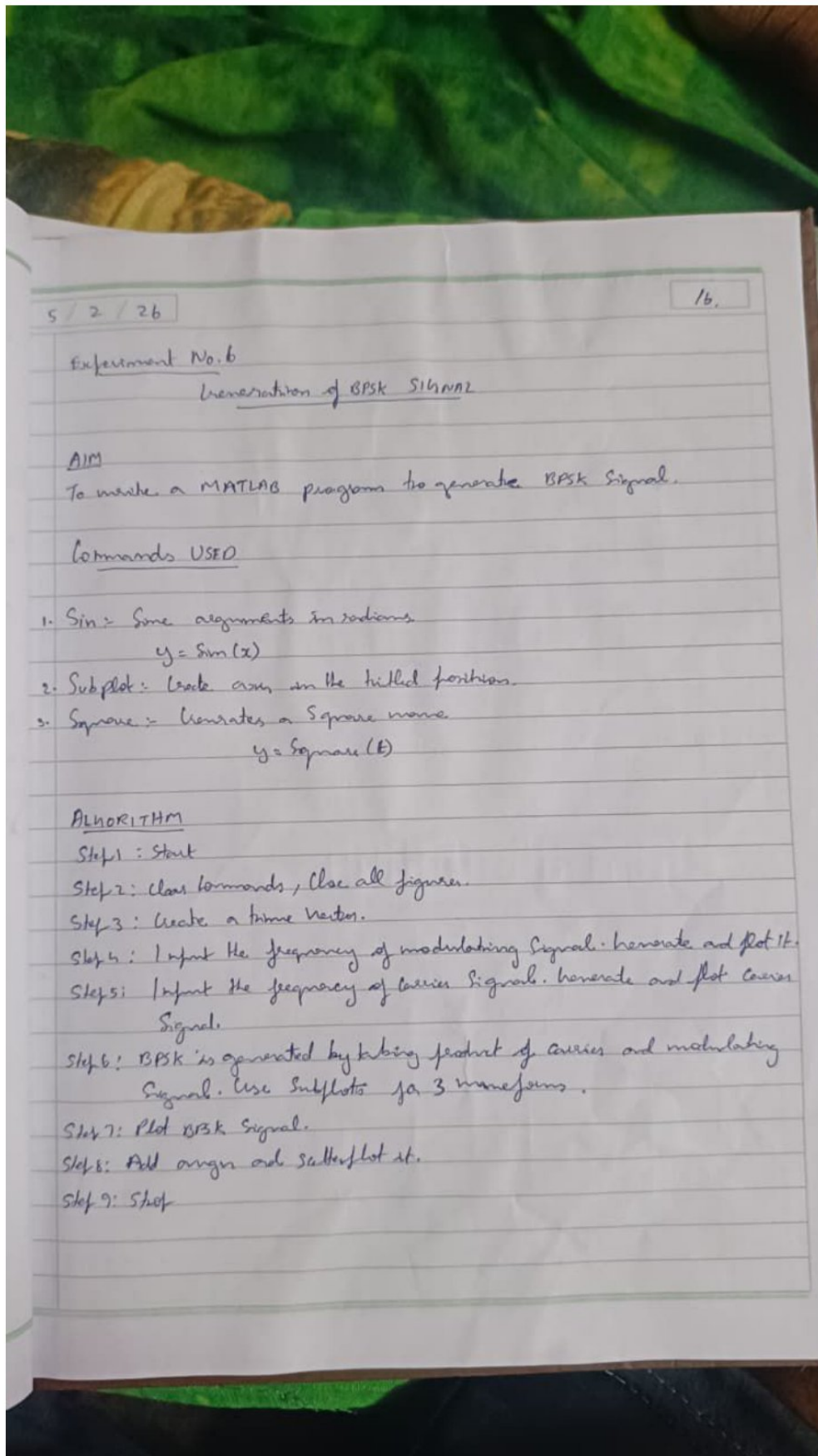




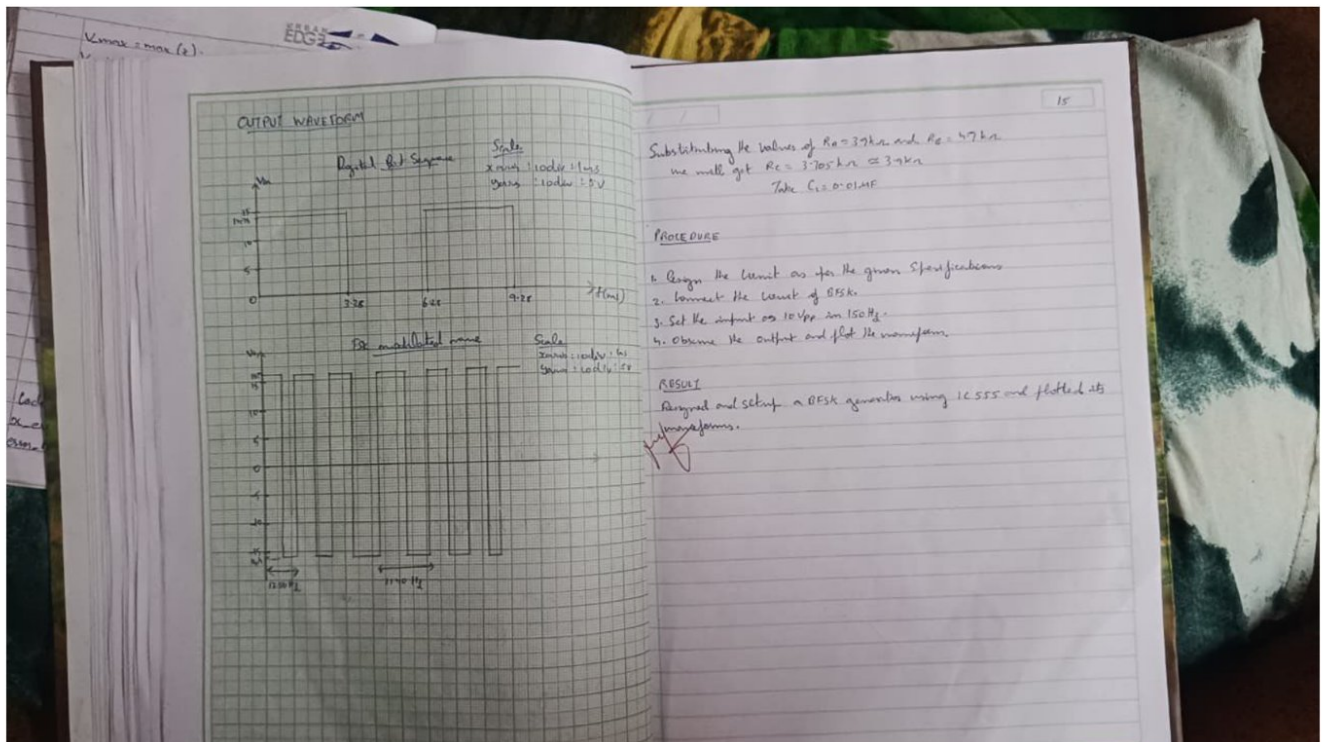


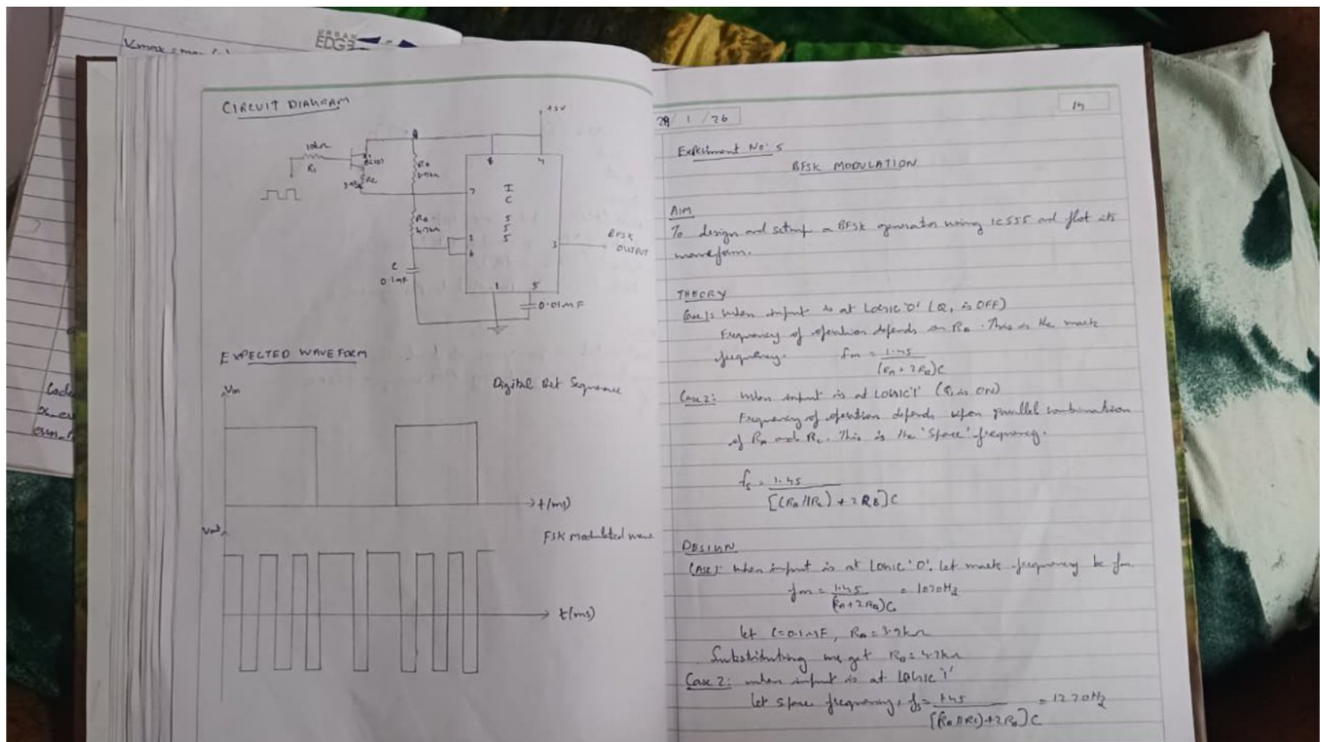


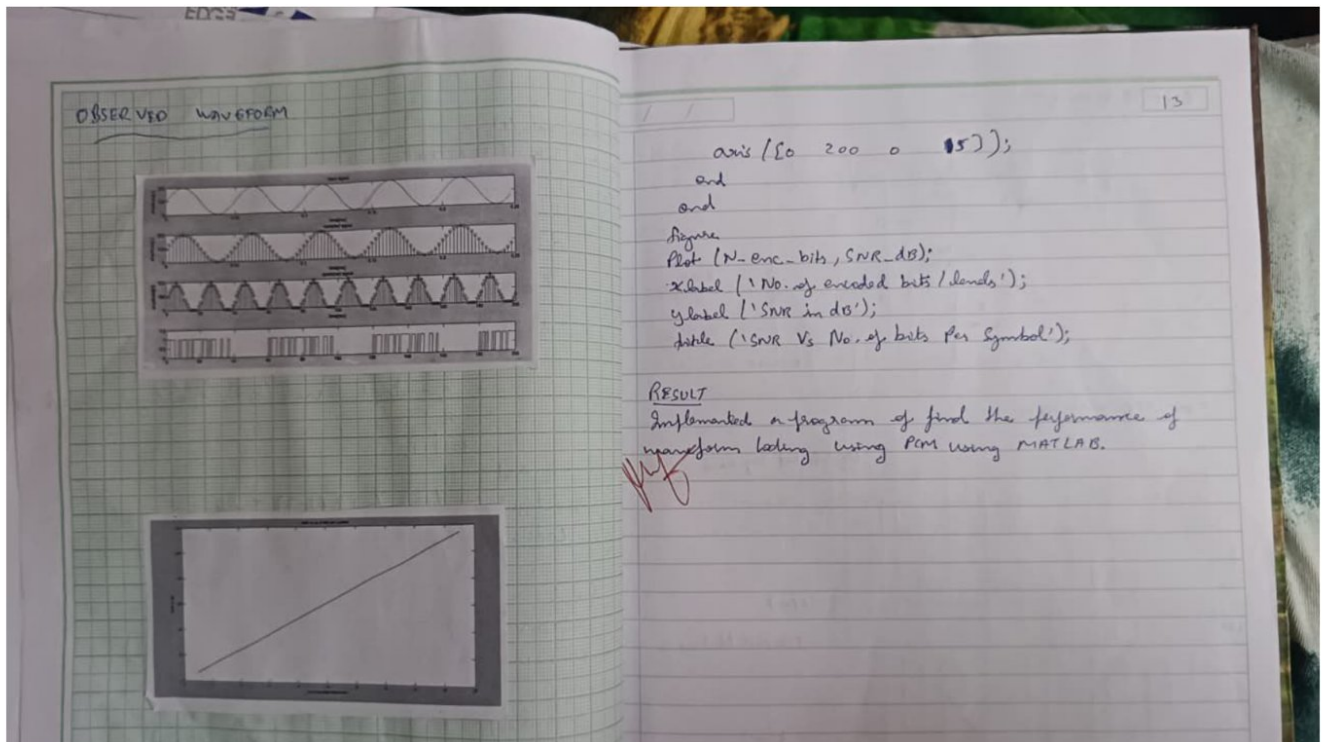


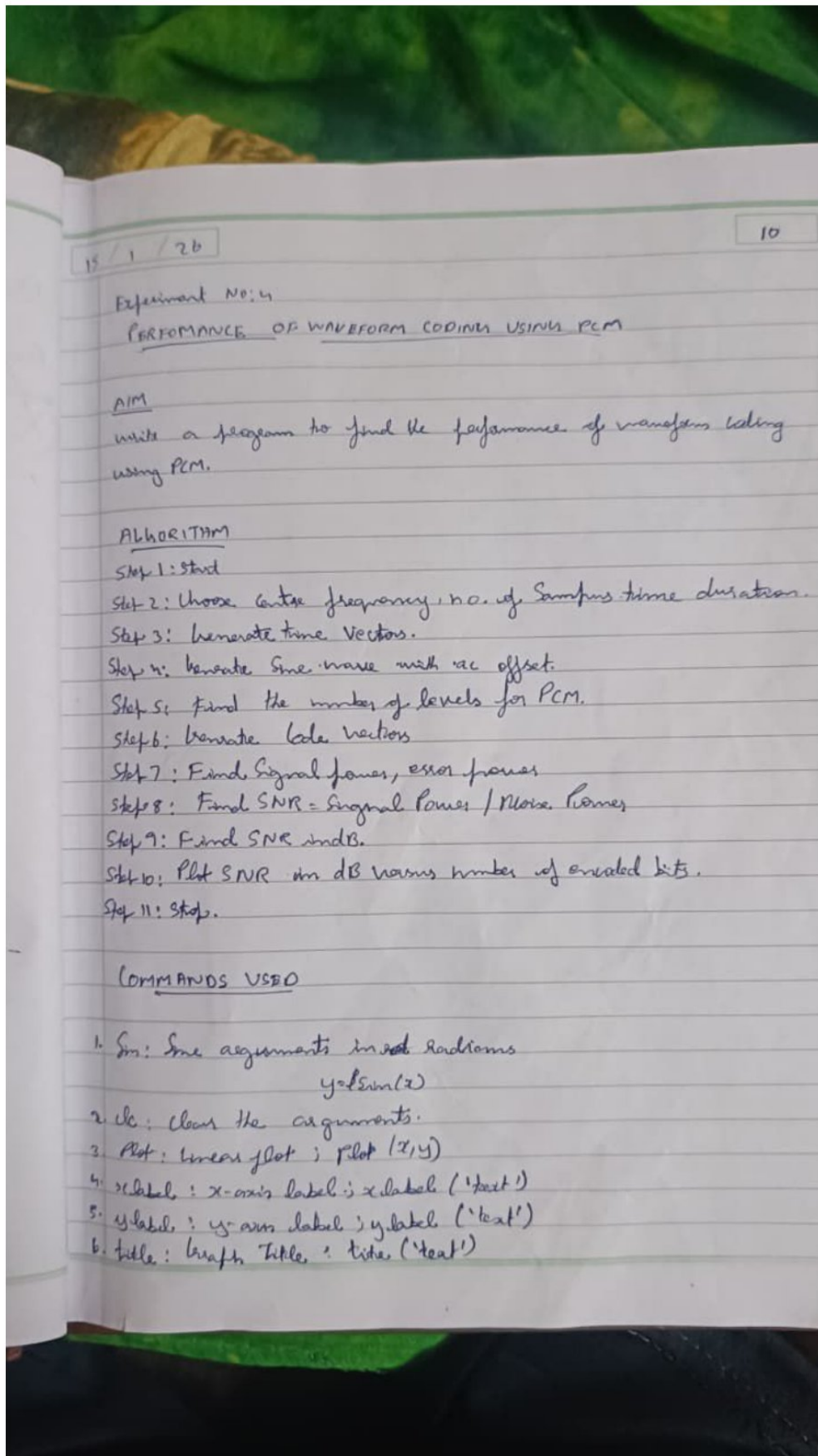


Original reference page 8

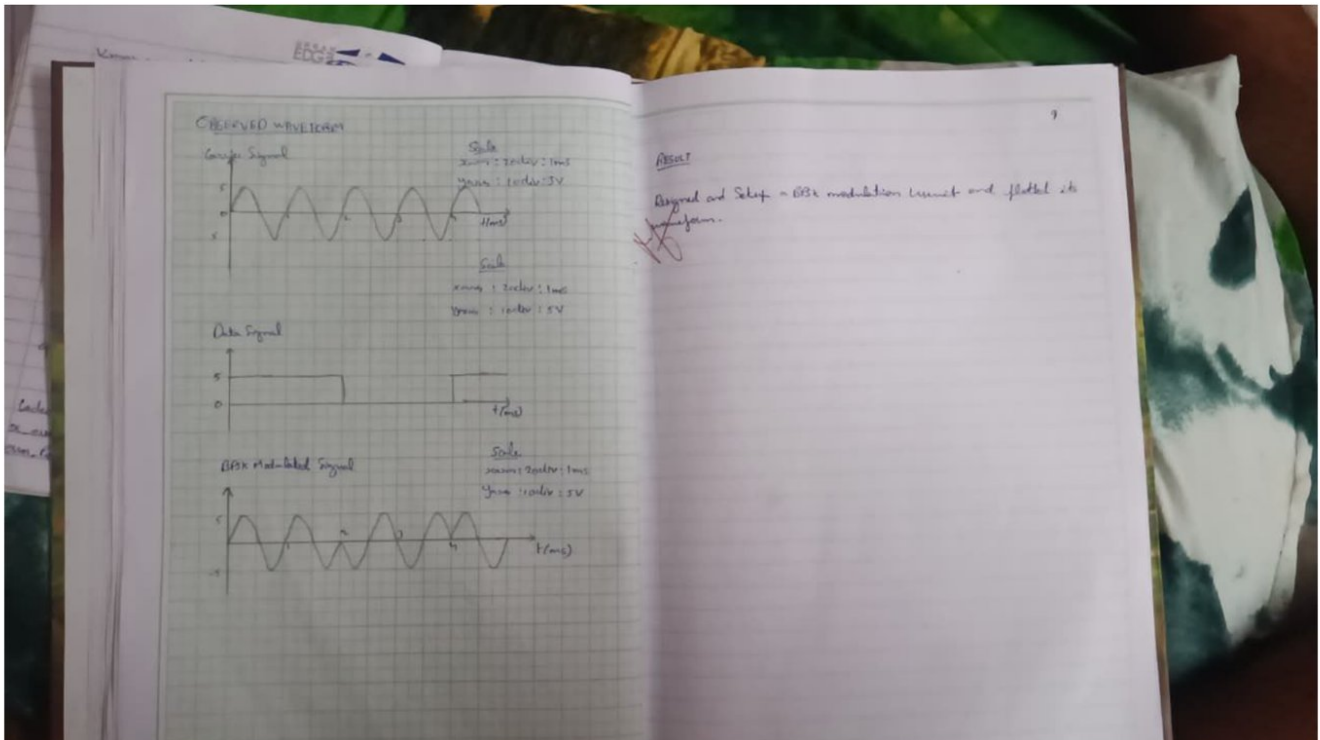


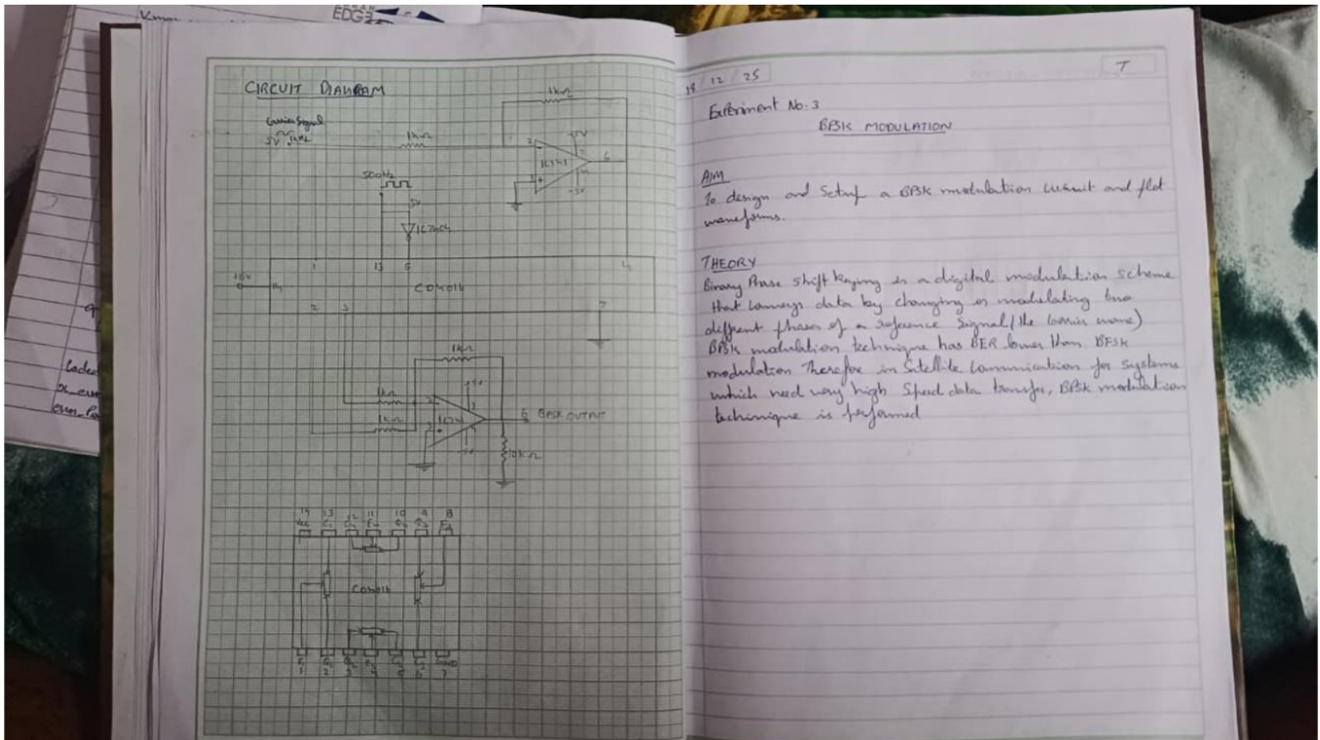


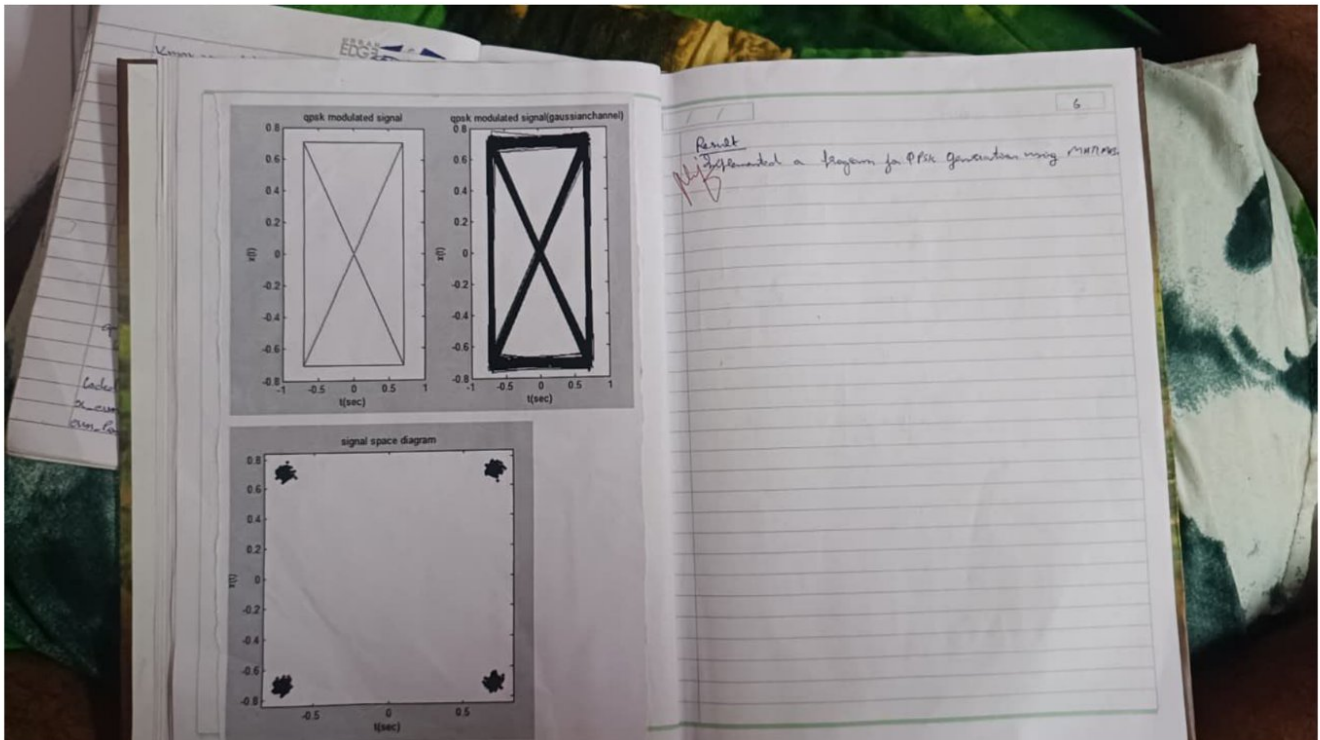


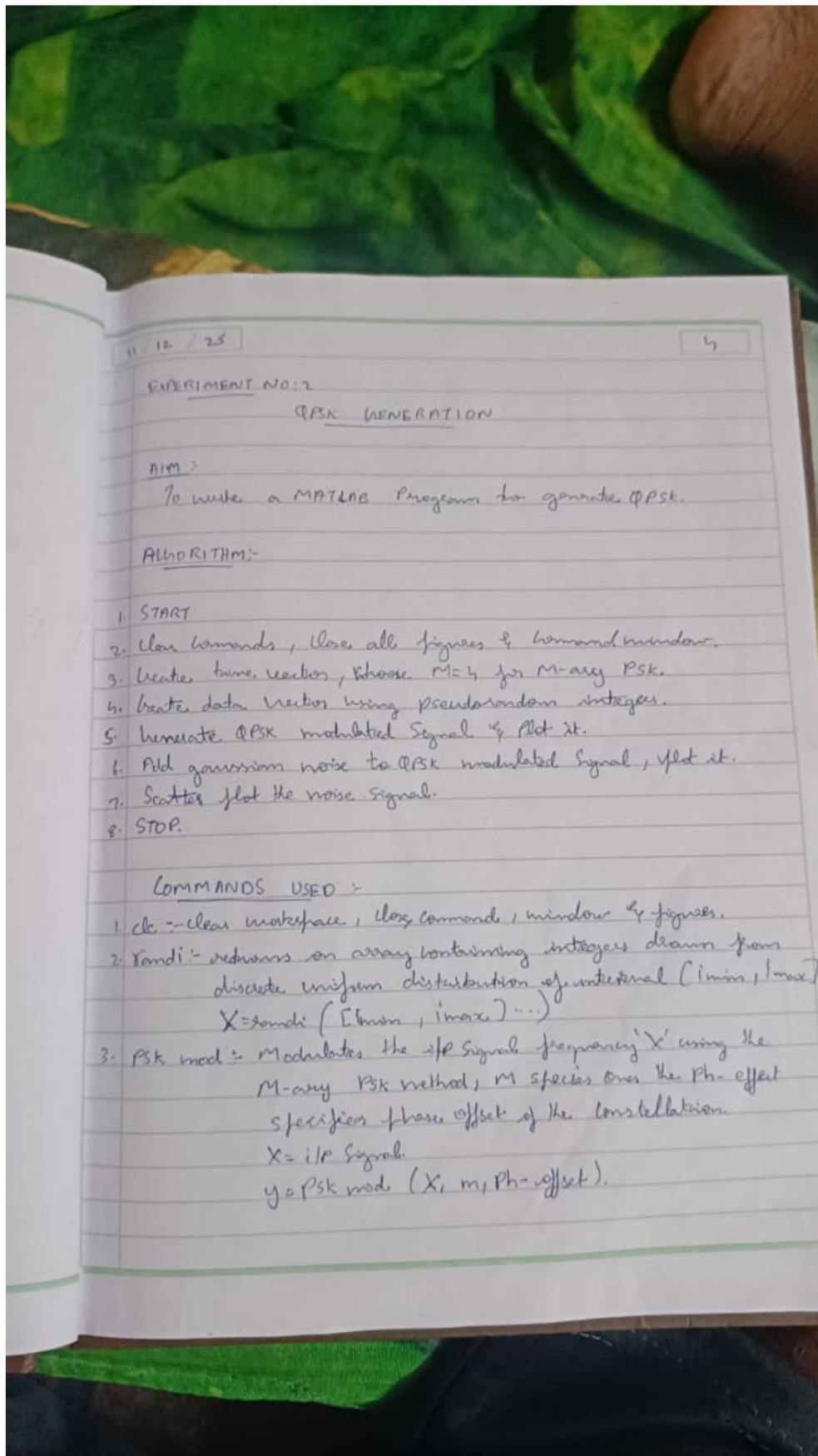


Original reference page 14









11/12/25

5

EXPERIMENT NO: 2

QPSK GENERATION

AIM:-

To write a MATLAB Program to generate QPSK.

ALGORITHM:-

1. START
2. Clear commands, close all figures & command window.
3. Create time vector, choose $M=4$ for M-ary PSK.
4. Create data vector using pseudorandom integers.
5. Generate QPSK modulated signal & plot it.
6. Add gaussian noise to QPSK modulated signal, plot it.
7. Scatter plot the noise signal.
8. STOP.

COMMANDS USED :-

1. `clc` :- clear workspace, clear commands, window & figures.
2. `randi` :- returns an array containing integers drawn from discrete uniform distribution of interval $[imin, imax]$
 $X = \text{randi}([imin, imax] \dots)$
3. `PSK mod` :- Modulates the i/p signal frequency 'X' using the M-ary PSK method; M specifies order the Ph-effect specifies phase offset of the constellation.
 $X = \text{i/p signal}$
 $y = \text{PSK mod}(X, m, \text{Ph-offset})$

